Anomaly Detection vs. Supervised Learning ⚖️

When you have a dataset with very few positive examples ($y = 1$) and a large number of negative examples ($y = 0$), the choice between using Anomaly Detection or Supervised Learning can be subtle.

Here is the framework for deciding which to use.

1. Based on Number of Examples

The most immediate indicator is the volume of positive data available.

Anomaly Detection	Supervised Learning
Very small number of positive examples ($y = 1$). (e.g., 0 to 20 examples).	**Large** number of positive examples ($y = 1$).
Large number of negative examples ($y = 0$).	**Large** number of negative examples ($y = 0$).
The algorithm learns $p(x)$ using **only** the negative examples. The few positive examples are reserved for the Cross-Validation and Test sets.	The algorithm uses **both** positive and negative examples during training to learn the boundary between classes.

2. Based on the Nature of the "Anomaly"

The more important distinction lies in whether future anomalies will look like past anomalies.

Use Anomaly Detection When:

- There are many **different types** of anomalies.
- It is hard for an algorithm to learn what an anomaly "looks like" from a small set of positive examples.
- **Crucial:** Future anomalies may look **nothing like** the anomalies you have seen so far.
- **Method:** The algorithm builds a model of "Normal." Anything that deviates from "Normal" is flagged.

Use Supervised Learning When:

- There are enough positive examples for the algorithm to get a sense of what they look like.
- **Crucial:** Future positive examples are likely to be **similar** to the positive examples in the training set.

- **Method:** The algorithm learns a boundary that specifically separates positive examples from negative examples.

Real-World Examples

Application	Recommended Algorithm	Reason
Financial Fraud	**Anomaly Detection**	Hackers constantly invent **new** methods of fraud. A new attack might look completely different from a previous one.
Email Spam	**Supervised Learning**	Spam often follows predictable patterns (selling drugs, phishing links). Future spam usually resembles past spam.
Manufacturing (Unknown Defects)	**Anomaly Detection**	Example: Aircraft engines. A failure might happen for a brand new reason (a new type of crack or material failure) never seen before.
Manufacturing (Known Defects)	**Supervised Learning**	Example: Scratched smartphone screens. You have seen thousands of scratches; you just want to find *more of the same* defect.
Data Center Security	**Anomaly Detection**	Hackers find new exploits (Zero-day attacks). The machine behaves in a "weird" way that hasn't been defined yet.
Weather / Medical	**Supervised Learning**	Diagnosing a specific disease or predicting rain vs. sun. The "positive" cases (having the disease/rain) are well-defined and repetitive.

Summary

- **Anomaly Detection:** "I don't know what might go wrong, but I know what 'normal' looks like. Alert me if *anything* weird happens."
- **Supervised Learning:** "I know exactly what goes wrong (or what I'm looking for). Here are 1,000 examples of it. Find me more of *this specific thing*."